

First 10 Observations of Dataset

Monday, 20 October, 2025 08:56:51 PM 1

Obs	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
1	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
2	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
3	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
4	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
5	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
6	57	1	0	140	192	0	1	148	0	0.4	1	0	1	1
7	56	0	1	140	294	0	0	153	0	1.3	1	0	2	1
8	44	1	1	120	263	0	1	173	0	0	2	0	3	1
9	52	1	2	172	199	1	1	162	0	0.5	2	0	3	1
10	57	1	2	150	168	0	1	174	0	1.6	2	0	2	1

The CONTENTS Procedure

Data Set Name	WORK.HEART	Observations	303
Member Type	DATA	Variables	14
Engine	V9	Indexes	0
Created	2025-10-20 20:56:51	Observation Length	112
Last Modified	2025-10-20 20:56:51	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	1168
Obs in First Data Page	303
Number of Data Set Repairs	0
Filename	/saswork/SAS_work3181000190C6_odaws02-usw2.oda.sas.com/SAS_work24A8000190C6_odaws02-usw2.oda.sas.com/heart.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	1074848683
Access Permission	rw-r--r--
Owner Name	u64371103
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
1	age	Num	8	BEST12.	BEST32.
12	ca	Num	8	BEST12.	BEST32.
5	chol	Num	8	BEST12.	BEST32.
3	cp	Num	8	BEST12.	BEST32.
9	exang	Num	8	BEST12.	BEST32.
6	fbs	Num	8	BEST12.	BEST32.
10	oldpeak	Num	8	BEST12.	BEST32.

The CONTENTS Procedure

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
7	restecg	Num	8	BEST12.	BEST32.
2	sex	Num	8	BEST12.	BEST32.
11	slope	Num	8	BEST12.	BEST32.
14	target	Num	8	BEST12.	BEST32.
13	thal	Num	8	BEST12.	BEST32.
8	thalach	Num	8	BEST12.	BEST32.
4	trestbps	Num	8	BEST12.	BEST32.

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
age	303	54.3663366	9.0821010	29.0000000	77.0000000
sex	303	0.6831683	0.4660108	0	1.0000000
cp	303	0.9669967	1.0320525	0	3.0000000
trestbps	303	131.6237624	17.5381428	94.0000000	200.0000000
chol	303	246.2640264	51.8307510	126.0000000	564.0000000
fbs	303	0.1485149	0.3561979	0	1.0000000
restecg	303	0.5280528	0.5258596	0	2.0000000
thalach	303	149.6468647	22.9051611	71.0000000	202.0000000
exang	303	0.3267327	0.4697945	0	1.0000000
oldpeak	303	1.0396040	1.1610750	0	6.2000000
slope	303	1.3993399	0.6162261	0	2.0000000
ca	303	0.7293729	1.0226064	0	4.0000000
thal	303	2.3135314	0.6122765	0	3.0000000
target	303	0.5445545	0.4988348	0	1.0000000

First 10 Observations of Dataset

Monday, 20 October, 2025 08:56:51 PM 5

Obs	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
1	0.9506240215	1	3	145	-0.255910365	1	0	150	0	2.3	0	0	1	1
2	-1.912149695	1	2	130	0.0720802521	0	1	187	0	3.5	0	0	2	1
3	-1.471722969	0	1	130	-0.815423771	0	0	172	0	1.4	2	0	2	1
4	0.1798772518	1	1	120	-0.198029668	0	1	178	0	0.8	2	0	2	1
5	0.2899839332	0	0	120	2.078611086	0	1	163	1	0.6	2	0	2	1
6	0.2899839332	1	0	140	-1.046946559	0	1	148	0	0.4	1	0	1	1
7	0.1798772518	0	1	140	0.9209971433	0	0	153	0	1.3	1	0	2	1
8	-1.141402925	1	1	120	0.3228966063	0	1	173	0	0	2	0	3	1
9	-0.260549474	1	2	172	-0.911891599	1	1	162	0	0.5	2	0	3	1
10	0.2899839332	1	2	150	-1.509992136	0	1	174	0	1.6	2	0	2	1

The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=2 Maxiter=100 Converge=0.02

Convergence criterion is satisfied.

Criterion Based on Final Seeds = 0.7993

Cluster Summary						
Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	125	0.8263	5.3544		2	1.7242
2	178	0.7818	2.6334		1	1.7242

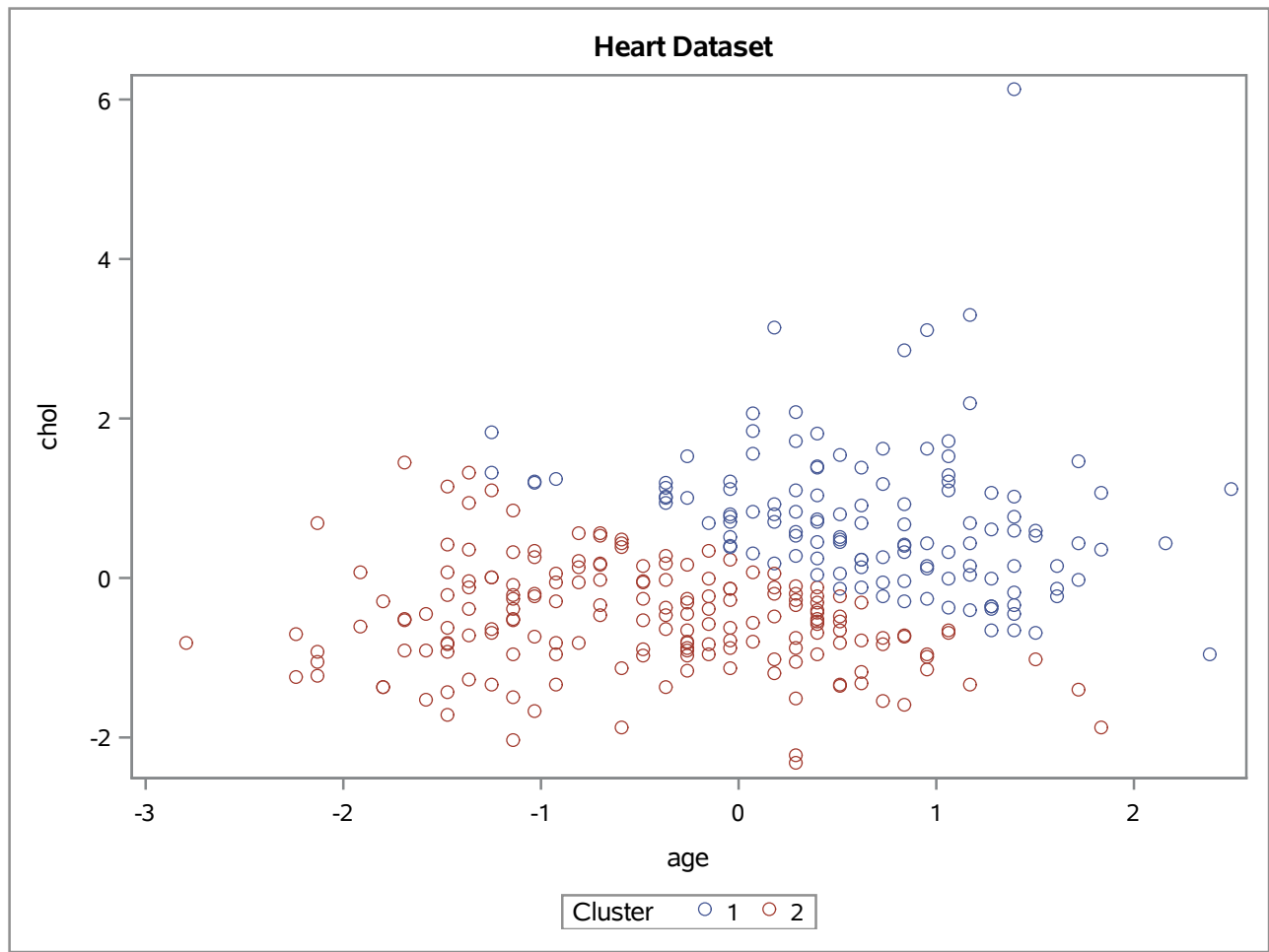
Pseudo F Statistic = 170.38

Observed Over-All R-Squared = 0.36145

Approximate Expected Over-All R-Squared = 0.37918

Cubic Clustering Criterion = -1.106

WARNING: The two values above are invalid for correlated variables.



The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=3 Maxiter=100 Converge=0.02

Convergence criterion is satisfied.

Criterion Based on Final Seeds = 0.6514

Cluster Summary						
Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	125	0.5824	2.0146		3	1.7225
2	109	0.6307	1.9761		1	1.7517
3	69	0.7999	4.8895		1	1.7225

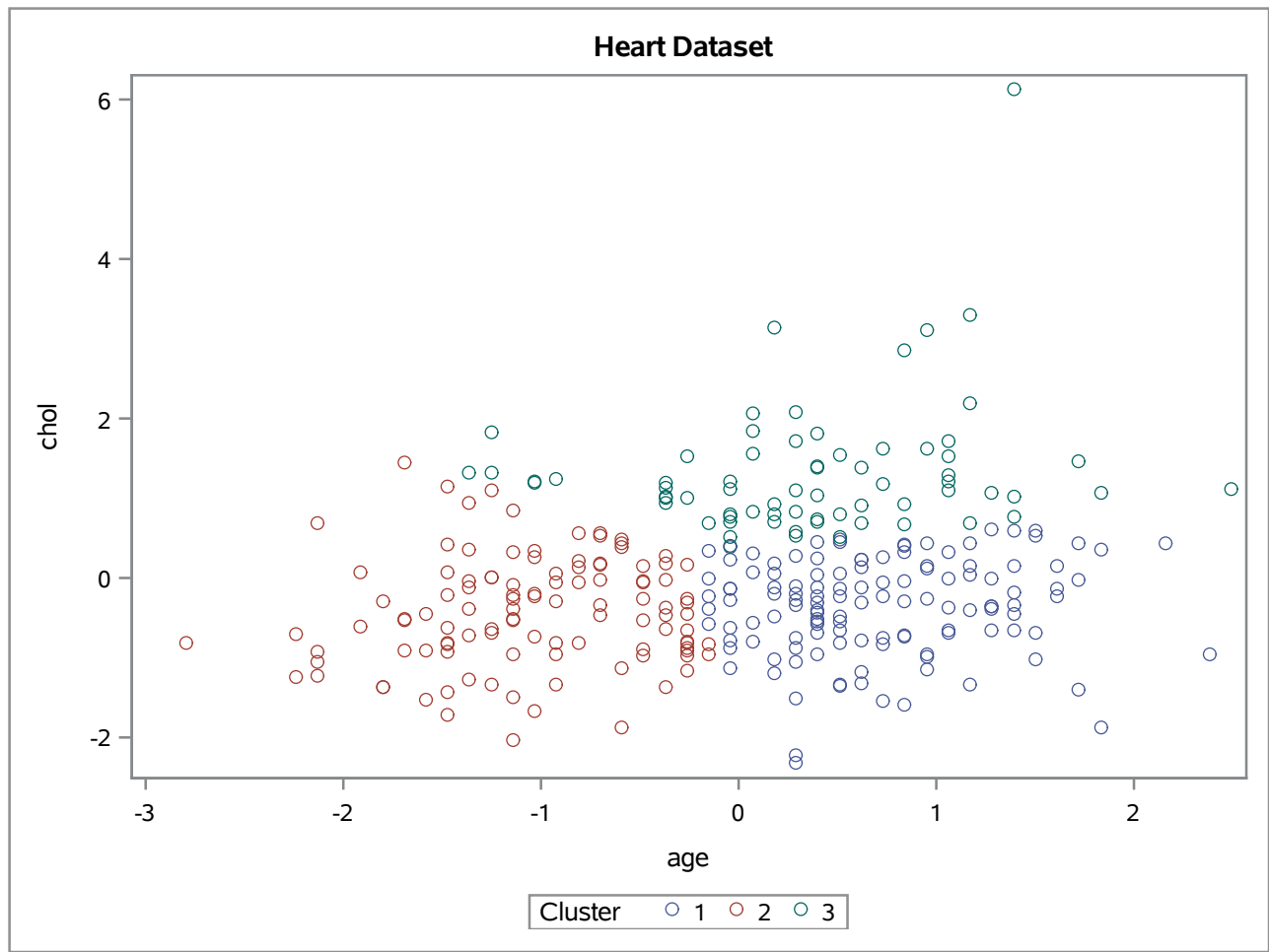
Pseudo F Statistic = 202.44

Observed Over-All R-Squared = 0.57439

Approximate Expected Over-All R-Squared = 0.67080

Cubic Clustering Criterion = -7.207

WARNING: The two values above are invalid for correlated variables.



The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=4 Maxiter=100 Converge=0.02

Convergence criterion is satisfied.

Criterion Based on Final Seeds = 0.6183

Cluster Summary						
Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	95	0.6570	2.4534		3	1.5823
2	97	0.6624	2.0506		3	1.7216
3	110	0.5503	1.9457		1	1.5823
4	1	.	0		1	5.2603

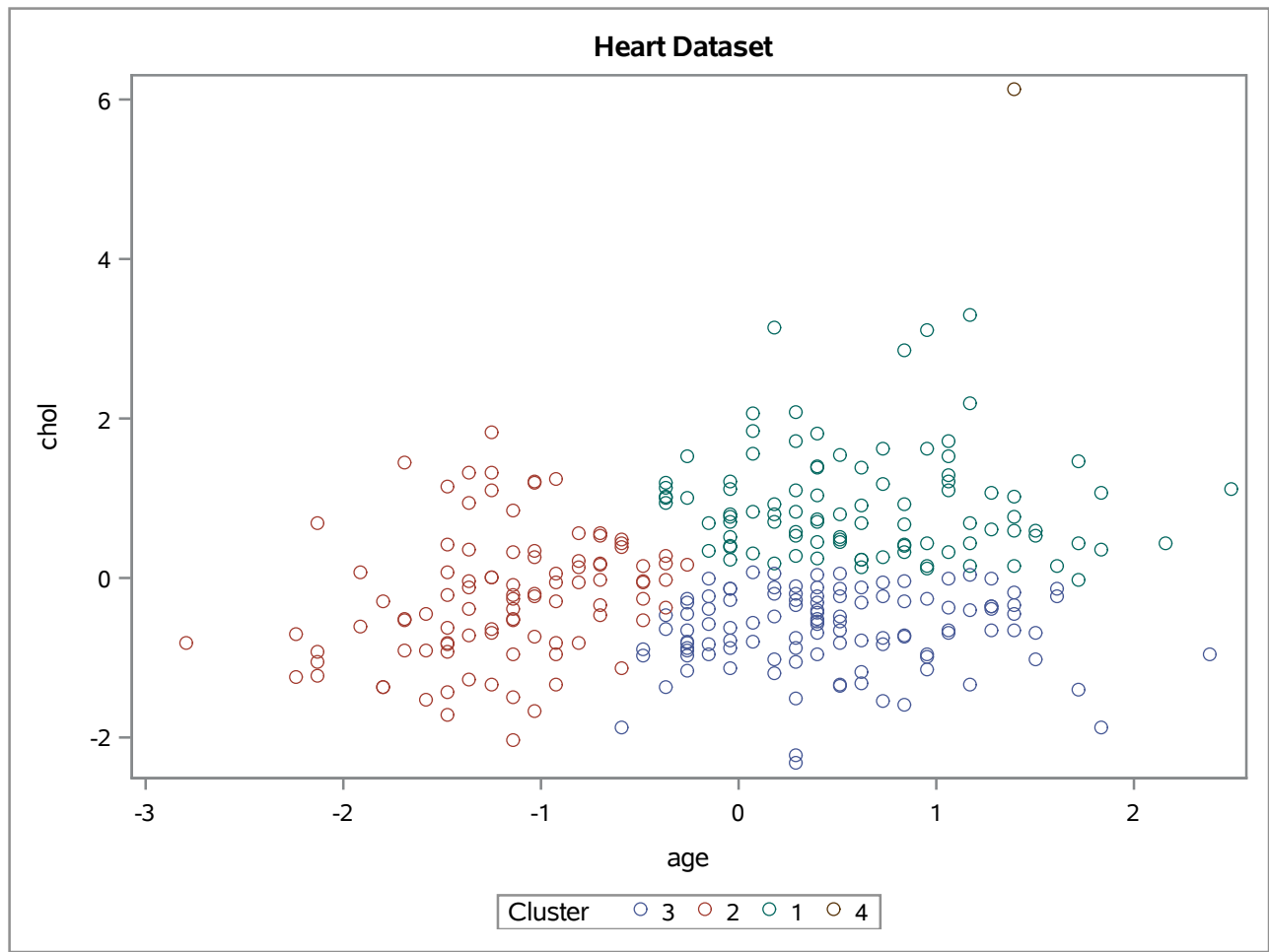
Pseudo F Statistic = 160.46

Observed Over-All R-Squared = 0.61686

Approximate Expected Over-All R-Squared = 0.75496

Cubic Clustering Criterion = -10.885

WARNING: The two values above are invalid for correlated variables.



The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=5 Maxiter=100 Converge=0.02

Convergence criterion is satisfied.

Criterion Based on Final Seeds = 0.5593

Cluster Summary						
Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	1	.	0		5	4.9802
2	88	0.6134	1.8121		4	1.4980
3	74	0.4992	1.8128		4	1.2208
4	80	0.4870	1.9376		3	1.2208
5	60	0.6490	2.1687		3	1.4550

Pseudo F Statistic = 163.54

Observed Over-All R-Squared = 0.68703

Approximate Expected Over-All R-Squared = 0.80543

Cubic Clustering Criterion = -10.710

WARNING: The two values above are invalid for correlated variables.

